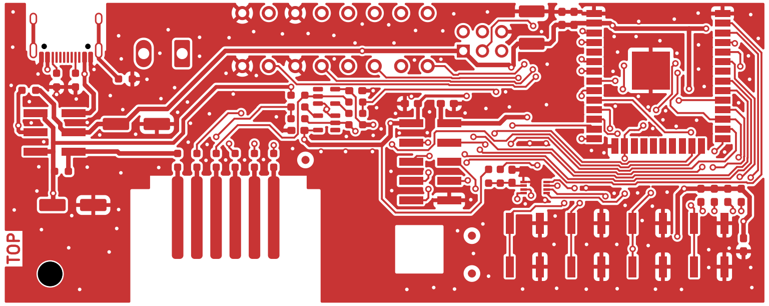
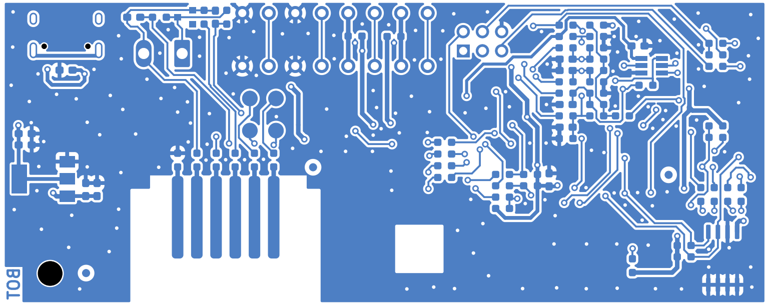
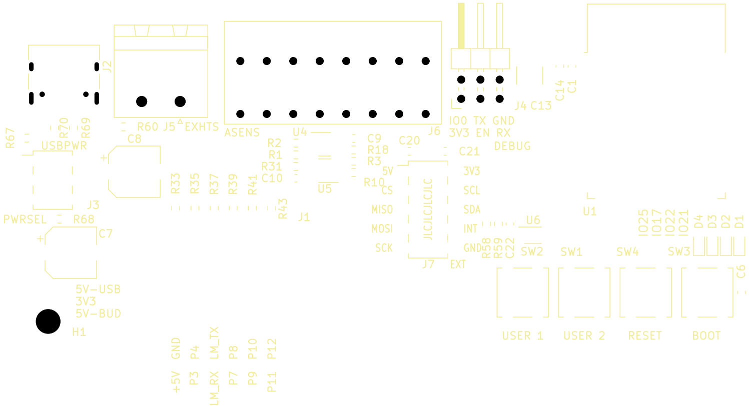
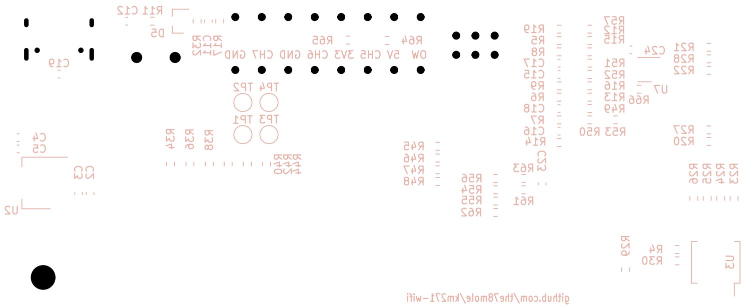
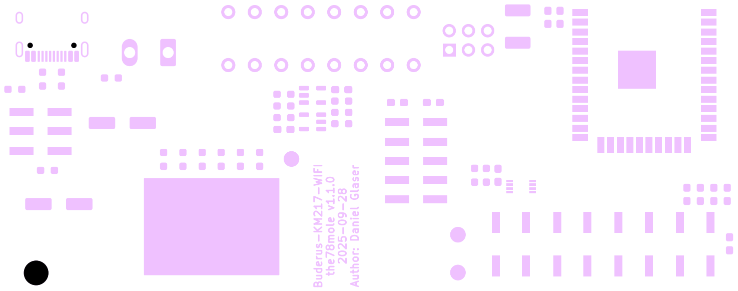

TOP

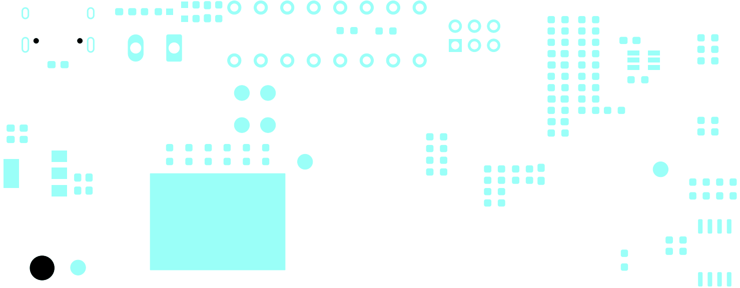


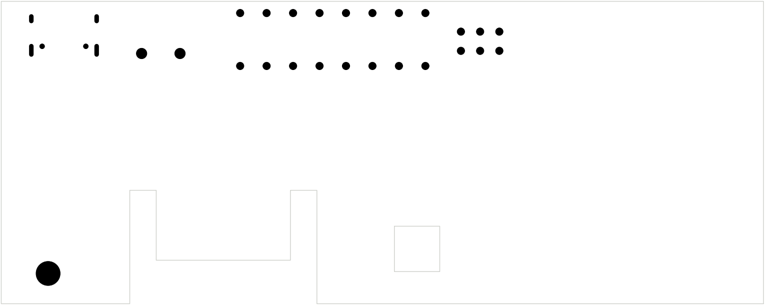
BOT

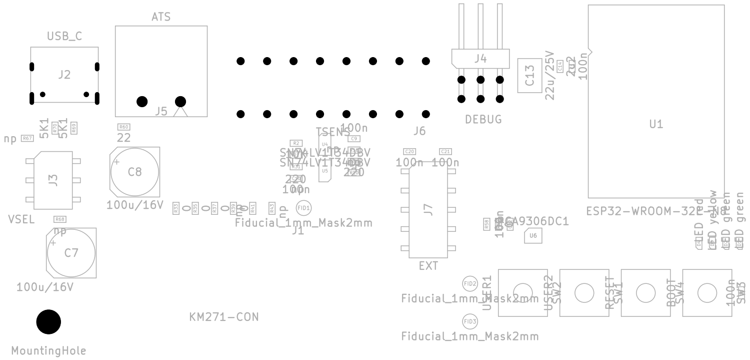

J2
R67
R70
R69
USBPWR
R60
J5
EXHTS
C8
ASENS
U4
R2
R1
R31
C10
C9
R18
R3
R10
U5
C20
J6
IO0
TX
GND
3V3
EN
RX
DEBUG
J4
C13
C14
C1
C21
5V
3V3
CS
SCL
SDA
MISO
INT
MOSI
GND
SCK
JLCJLCJLCJLC
J7
EXT
R58
R59
C22
U6
U1
IO25
IO17
IO22
IO21
D4
D3
D2
D1
SW2
SW1
SW4
SW3
C6
USER 1
USER 2
RESET
BOOT
R33
R35
R37
R39
R41
R43
J1
J3
PWRSEL
R68
C7
5V-USB
3V3
5V-BUO
H1
+5V
GND
P3
P4
LM_RX
LM_TX
P7
P8
P9
P10
P11
P12

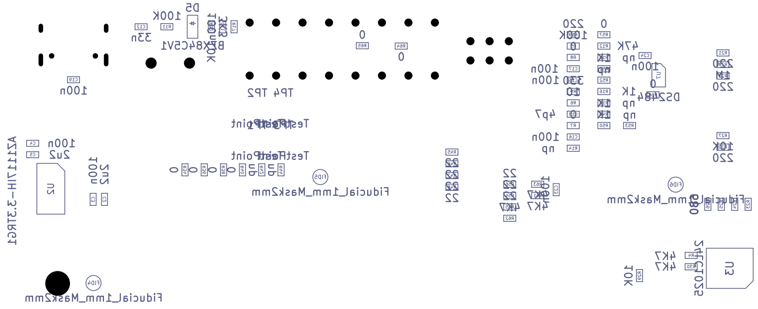

GND CH7 GND CH6 3V3 CH5 5V OW
TP1 TP2 TP3 TP4
U2 U3 U7 D5
github.com/the78mole/km271-wifi

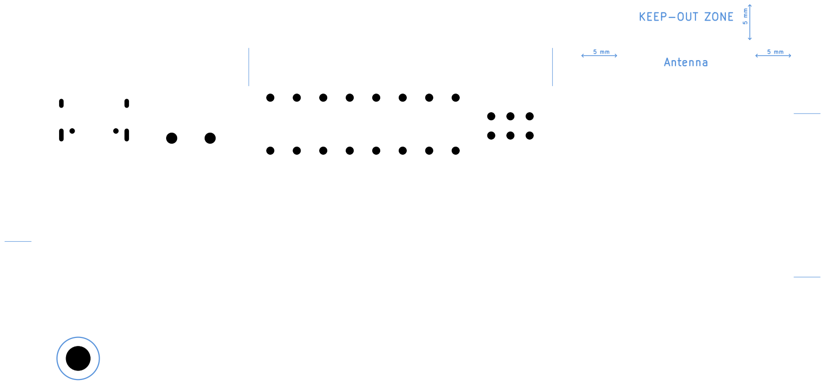

Buderus-KM217-WIFI
the78mole v1.1.0
2025-09-28
Author: Daniel Glaser






ATS
USB_C
J2
J5
J4
C13
22u/25V
2u2
100n
DEBUG
U1
J6
TSENS
100n
5K1
5K1
np
22
J3
C8
VSEL
np
C7
100u/16V
100u/16V
SN74LV1T34DBV
SN74LV1T34DBV
10K
220
220
100n
100n
100n
J7
EXT
0
0
0
np
np
Fiducial_1mm_Mask2mm
J1
ESP32-WROOM-32E-N8
PCA9306DC1
U6
100n
LED red
LED yellow
LED green
LED green
USER1
USER2
SW2
RESET
SW1
BOOT
SW4
100n
SW3
Fiducial_1mm_Mask2mm
Fiducial_1mm_Mask2mm
KM271-CON
MountingHole


D5
100K
33n
BLK84C5V1
100n
TP4 TP2
TestPoint
TestPoint
Fiducial_1mm_Mask2mm
Fiducial_1mm_Mask2mm
Fiducial_1mm_Mask2mm
AZ1117IH-3.3TRG1
U2
100n
2u2
DS2484
47K
220
100K
4p7
10K
220
1M
U3
22
4K7
10K



FAB NOTE J1 (KM271 card edge): 12 gold fingers F+B, 1.6 mm FR4 (+-10%).
Order option: Gold Fingers = YES (ENIG / hard gold on fingers), mask open over fingers.
Bevel: NOT POSSIBLE - finger edge (y=124.25) is recessed 5.75 mm between the board legs (y=130) and JLC chamfers only boards >= 50x50 mm. Original module is used unbevelled.
Fingers end 0.2 mm before the board edge; no vias/pads within 1 mm (rule area GOLD_FINGERS).

FAB NOTES Buderus-KM217-WiFi rev v1.1.0
Stackup: 2 layers, FR4 TG135, 1.6 mm +-10 pct, outer copper 1 oz (35 um)
L1 F.Cu signal/GND pour | core 1.5 mm | L2 B.Cu signal/GND pour
Finish: ENIG (gold fingers). Gold fingers: YES, bevel: NO (see J1 note)
Mask green, silk white, vias tented. Min track/space 0.20/0.20 mm, min via 0.3/0.6 mm
Assembly: Standard PCBA, both sides. BOM/CPL: production/bom.csv, production/positions.csv